AI-based Urban Traffic Management System that uses Reinforcement Learning and Computer Vision

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PROJECT GUIDE
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MOTIVATION

SRM INSTITUTE OF SCIENCE & TECHNOLOGY (Deemed to be University u/s 3 of UGC Act, 1956)

- Rapid increase in urban traffic congestion leading to higher travel time and accident risks
- Emergency vehicles face critical delays during peak traffic, affecting response time and public safety
- Traditional fixed-time traffic signals fail to adapt to dynamic conditions like traffic surges, pedestrians, and weather
- Growing need for intelligent, real-time, and energy-efficient traffic control systems
- Lack of systems that simultaneously consider traffic, pedestrians, weather, emergency priority, and energy optimization
- Need for low-cost, scalable, and real-world deployable solutions for next-generation urban mobility

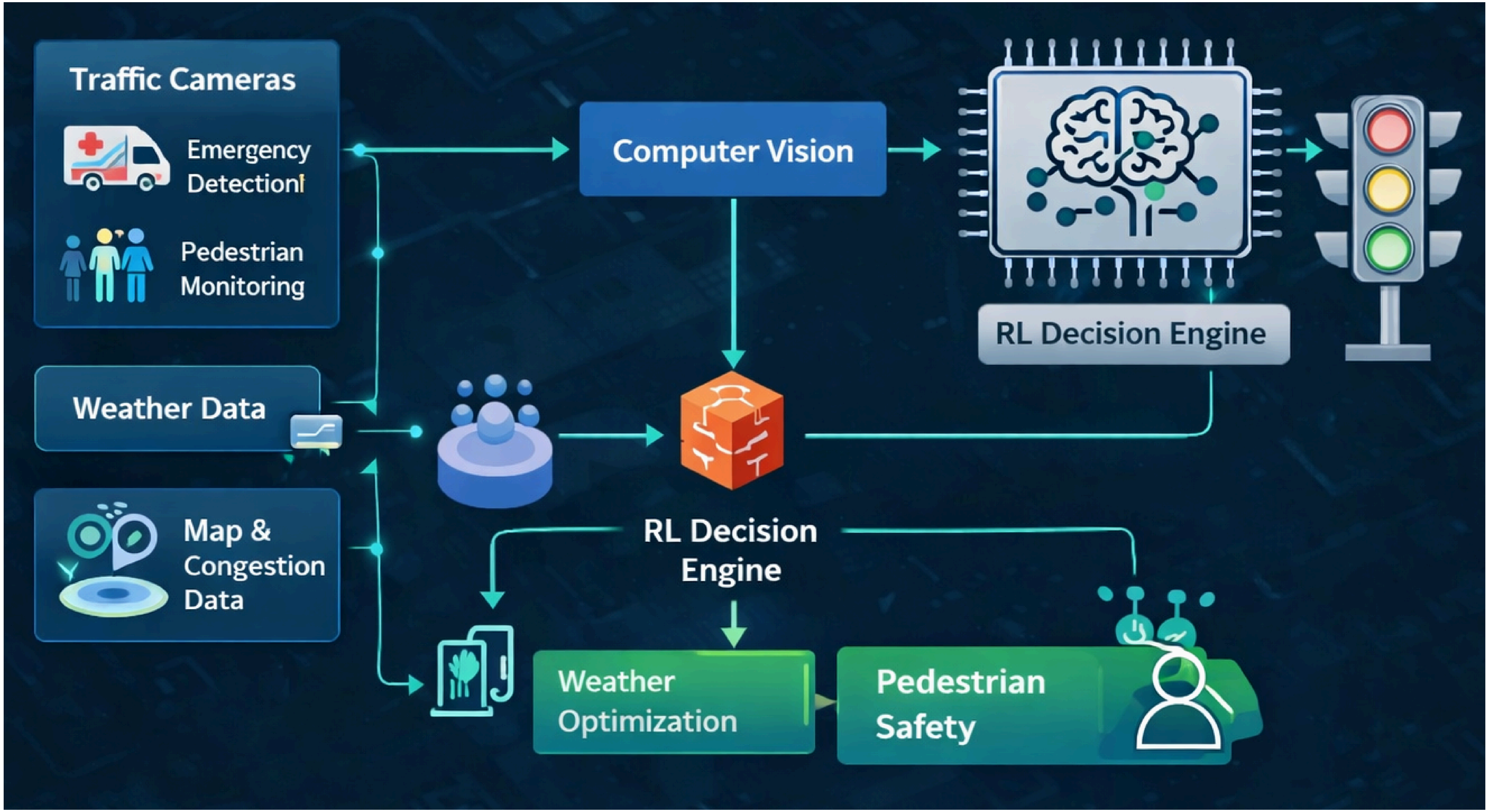
PROBLEM STATEMENT

Urban traffic systems often struggle to manage changing conditions such as varying traffic density, weather changes, pedestrian movement, and emergency vehicle needs. Traditional traffic signals do not adapt in real time, which leads to congestion, longer delays, safety risks, and increased energy consumption. Therefore, there is a need for an intelligent and adaptive traffic management system that can effectively respond to real-time road and environmental conditions.

OBJECTIVE

- Develop adaptive traffic signal using RL + CV
- Enable real-time emergency vehicle priority
- Ensure pedestrian safety via vision monitoring
- Optimize traffic using live congestion & weather data
- Reduce fuel waste and energy consumption

METHODOLOGY



TOOLS REQUIRED

HARDWARE

- Raspberry Pi 4
- Traffic Light LED Module
- Camera
- Ultrasonic
- Breadboard & Jumper Wires
- Power Supply

SOFTWARE

- Raspberry Pi OS
- Python
- TensorFlow
- OpenCV
- YOLOv8
- GPIO Libraries

PLANNED SCHEDULE

Phase	Milestone	Date
1	Team Formation & Topic Decided	5 Jan 2026
2	Work Distribution	12 Jan 2026
3	Literature Survey Completed	19 Jan 2026
4	Abstract Preparation	27 Jan 2026
5	0th Review (Synopsis)	9 Feb 2026
6	1st Review (Implementation)	23 Feb 2026*
7	2nd Review (Testing)	16 Mar 2026*
8	3rd Review (Final Demo)	6 Apr 2026*

CONCLUSION

An intelligent urban traffic management system using Reinforcement Learning (RL) and Computer Vision (CV) is proposed to enhance traffic signal control. The system detects emergency vehicles, monitors pedestrian movement, considers weather conditions, and uses real-time congestion data for adaptive signal decisions. It dynamically adjusts signal timing based on changing traffic and environmental conditions. The system improves traffic flow, reduces delays for emergency vehicles, increases pedestrian safety, and lowers energy consumption compared to traditional traffic signal systems.

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THANK YOU!